

CSCI3240 Introduction to Computer Vision

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Lab 01 - Image basics (numpy)

Goal: get comfortable treating images as numpy arrays.

Setup (used by every task; no image files are needed - build the sample image in code):

```
import numpy as np
H, W = 64, 64
img = np.zeros((H, W, 3), dtype=np.uint8)
img[..., 0] = np.linspace(0, 255, W).astype(np.uint8)[None, :] # R: left-to-right gradient
img[..., 1] = np.linspace(0, 255, H).astype(np.uint8)[:, None] # G: top-to-bottom gradient
img[..., 2] = 128 # B: constant
```

Task 1. Build the 64x64 RGB gradient image above with numpy. Print its shape, dtype, min, max and the mean of each channel (expected means approx. 127.5, 127.5, 128.0).

Task 2. Convert it to grayscale with $\text{gray} = 0.299 R + 0.587 G + 0.114 B$ using numpy only (result dtype uint8). Print `gray[0, 0]`, `gray[63, 63]` and the mean gray level.

Task 3. Crop the central 32x32 region and flip it horizontally using array slicing only (no loops). Print the crop shape and check that flipping twice returns the original crop (`np.array_equal`).

Task 4. Compute the 256-bin intensity histogram of the grayscale image with `np.bincount(minlength=256)`. Print the number of non-empty bins and the five most frequent gray levels with their counts.

Task 5. Threshold the grayscale image at its mean intensity: `mask = gray > gray.mean()`. Print the foreground pixel count and ratio, then print the mask downsampled to 8x8 blocks as ASCII art ('#' where more than half of the block is foreground, '.' otherwise).

Deliverable: lab01.ipynb with one code cell per task; every cell must run and print its result.

Submit on WSP before Friday 23:59 (week 2). Only numpy is required.